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Current model capabilities and future predictive pathways of wildfires under a changing climate

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Wildland fire activity is expected to change substantially under future climate warming, with important implications for ecosystems, air quality, and climate feedbacks. In this Review, we compare land surface, Earth system, statistical, and machine learning model approaches that are used to project future fire activity across global and regional scales under multiple climate scenarios. Despite broad agreement that fire risk will increase in many extra-tropical regions, the magnitude of projected changes remains highly variable and uncertain due to differences in climate sensitivity, vegetation responses, human influences, and fire process representation. Recent advances include improved fire-process representation, increased coupling of fire with Earth system processes, and emerging frameworks that integrate observations, process-based modelling, and machine learning. By synthesising results across model types, we show that different modelling approaches provide complementary strengths for distinct applications, and that the greatest confidence in future wildfire projections arises where independent modelling approaches converge. Historical model evaluation and multi-model intercomparison projects provide a critical basis for interpreting disagreements among projections and identifying priorities for improving next-generation wildfire models.

Wildland fires (hereafter 'fires') are increasing in frequency, intensity and societal impact in many regions under climate warming[1–4]. Fires alter carbon cycling, vegetation structure and land-atmosphere feedbacks, with implications for future climate[5–7], ecosystems[8,9], air quality, human health[10–12], infrastructure and biodiversity through complex interactions between vegetation, weather, atmospheric composition and human activity. Fire emissions degrade air quality and contribute to global mortality (e.g. 1.53M deaths per year globally[13]), while suppression costs and damages continue to rise in many fire-prone regions (e.g. $3B per year in the United States[14]).

Modelling how fire regimes will evolve with climate change is increasingly important for mitigation and adaptation, yet fire modelling remains highly uncertain. The uncertainties arise from interacting processes, spanning vegetation dynamics, fuel availability and moisture, ignitions, meteorology, human activity, emissions, plume evolution, atmospheric chemistry and climate feedbacks[3,8,15]. Models differ substantially in how these processes are represented, ranging from process-based land surface and Earth system models to statistical and machine learning approaches. Consequently, future projections (out to the end of the century) of burned area, emissions, fire weather and smoke exposure vary widely across regions and climate scenarios.

Previous reviews have examined wildfire-climate interactions, and regional[16,17] or global fire trends[18,19]. However, rapid advances in modelling capability, Earth system coupling, observational constraints and machine learning now motivate a broader synthesis focused specifically on future fire projection pathways. Here, we review major modelling frameworks used to project future fire activity under climate change, including land surface models, Earth system models and statistical and machine learning approaches (see SI Note 1 for literature review methods). We examine how these frameworks represent fire processes and climate feedbacks, evaluate their historical performance and limitations, and synthesise their projections across regions, scenarios and modelling approaches. Despite the large uncertainties, many studies project increasing fire activity across northern extra-tropical and boreal regions. Our synthesis shows that the different modelling frameworks provide complementary strengths for distinct scientific applications, and that confidence in future projections is greatest where independent approaches converge and are supported by historical model evaluation. Finally, we outline several priorities for improving the next-generation fire-climate models.

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Mechanistic changes in fires under climate change

Wildfires interact with climate through complex processes that involve meteorology, vegetation and atmospheric composition. Major mechanisms linking climate change and wildfire activity are summarised and illustrated in Fig. 1.

Climate controls on wildland fires

Climate change influences wildfires through changes in meteorological conditions, fuel availability and ignition sources. Globally, moisture and air temperature are among the strongest climatic controls on fire occurrence as they contribute to fuel flammability[20–23], although dominant drivers vary regionally (Fig. 2). Even in regions where precipitation increases, warming-driven evaporation can still reduce fuel moisture and increase fire risk[23–26].

Earlier spring snowmelt lengthens the seasonal drying period[27–29] and permafrost thaw in the high north can increase soil drainage and upper-soil drying[26,30–34]. Climate-driven vegetation shifts further alter fuel structure and availability, and CO_2 fertilisation enhances biomass[35]. Biome transitions can either suppress or enhance fire activity. For example, woody encroachment in savannas may reduce burned area[36], whereas expanding taiga and increasing tree mortality in high-latitude forests may increase flammability[32,37–43].

Lightning is a climate-sensitive and spatially heterogeneous driver of fire ignition[44–49], and burned area in some remote regions is particularly sensitive to changes in lightning activity[3,20–23,50–54]. Additionally, winds and terrain influence fire spread[8], with atmospheric blocking events associated with persistent extreme fire weather[55,56].

While not a climate control, human activity can also influence fires via agricultural expansion, fire suppression, land fragmentation, accidental ignition and deforestation[57–65]. In some regions, increasing suppression efforts partially offset worsening fire weather[62–64].

When major climate forcers coincide, extreme fire weather can develop, increasing the likelihood of large and severe fires[23,29,66,67]. Such trends are already evident in boreal forests, the western U.S. and Australia[20,32,68], with attribution studies indicating a significant climate-change contribution[20,21,69–71].

Wildland fire controls on climate

Wildfires influence climate through changes to vegetation, atmospheric composition and chemistry (Fig. 1). The feedback can either amplify or dampen climate change depending on ecosystem type, burn severity, optical properties of smoke, atmospheric conditions and timescale.

Fires alter vegetation structure, ecosystem succession and land-atmosphere energy exchange[72,73]. Vegetation loss reduces evapotranspiration and latent cooling, often producing warmer and drier near-surface conditions[74]. Severe or frequently returning fires may additionally consume organic soils and alter ecosystem recovery pathways, promoting long-term biome shifts[37–39,75–77]. Such shifts may reduce flammability locally, but can weaken long-term carbon uptake and potentially convert some ecosystems from net carbon sinks to sources[78–83].

Fires modify atmospheric composition through emissions of greenhouse gases, aerosols and ozone precursors[84], influencing tropospheric ozone formation, modifying atmospheric oxidative capacity[84–86] and altering methane lifetime[86]. In peat-rich ecosystems, deep burning of organic soils additionally releases long-term carbon stores to the atmosphere[87].

Biomass burning (BB) aerosols, primarily black carbon and organic carbon, additionally influence climate through direct radiative effects and aerosol-cloud interactions. Their net forcing remains uncertain and depends strongly on injection height, aerosol properties, atmospheric conditions, transport and cloud interactions[88–98]. Through impacts on atmospheric circulation and precipitation, BB aerosols may influence

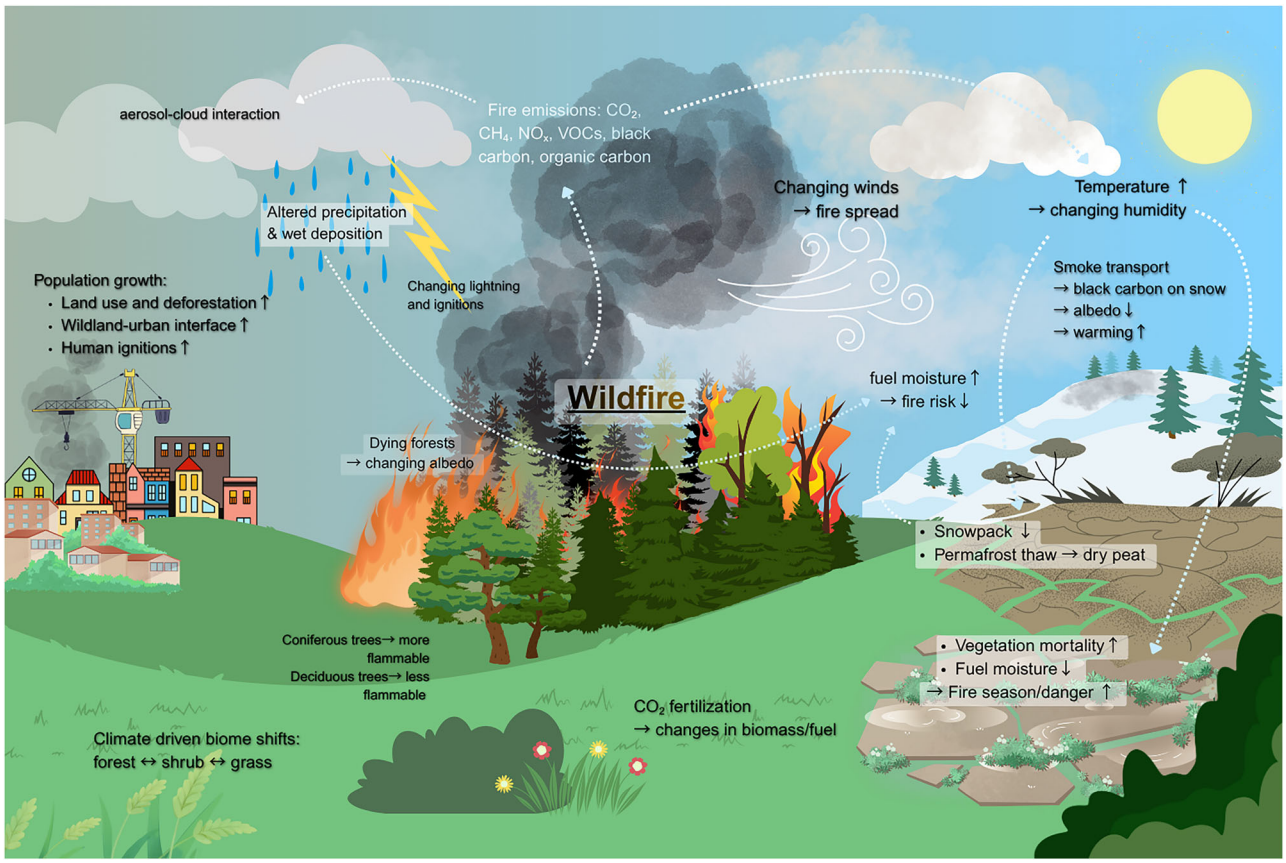


Fig. 1 | Climate-fire interactions. Schematic of the numerous factors and interactions between wildfires, climate and their feedbacks.

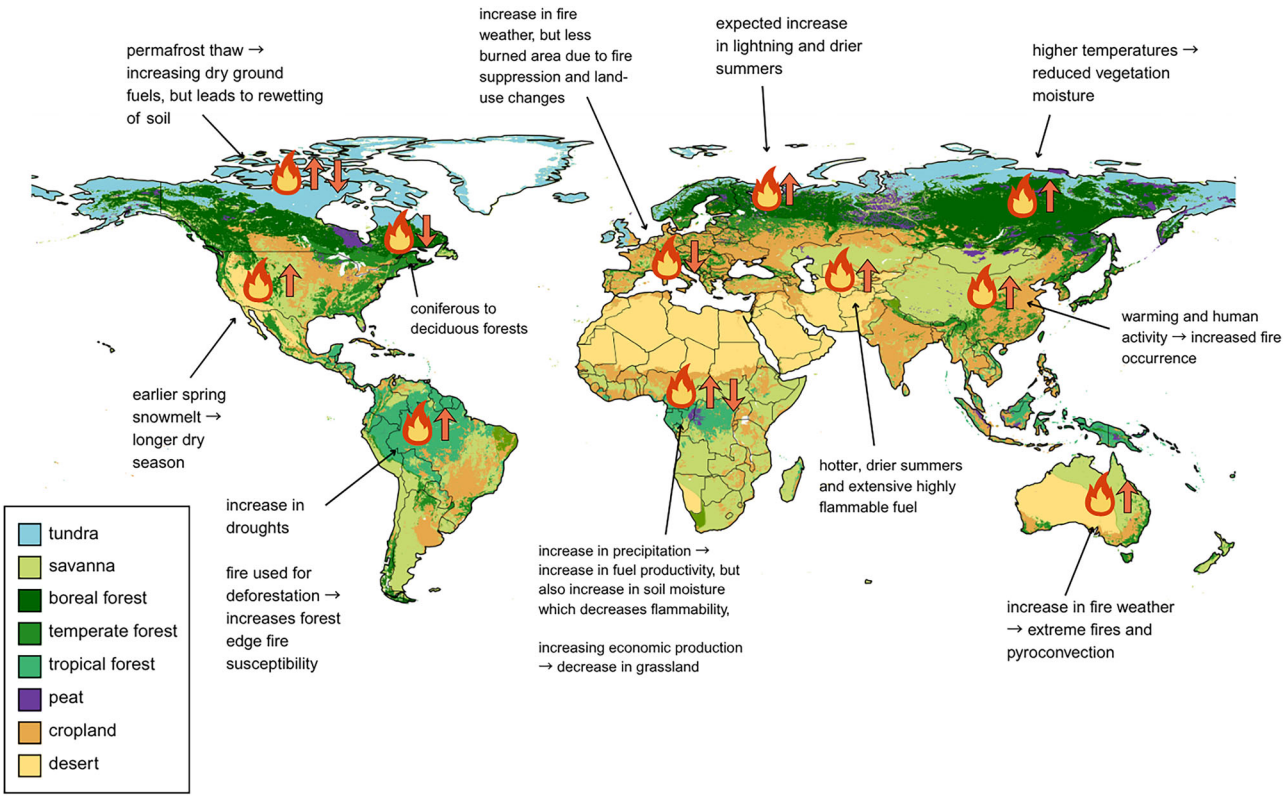


Fig. 2 | Global schematic of key regional wildfire drivers and how climate change is altering them. Background land cover types are from 2018 and derived from ESA CCI[278] and Xu et al.[279].

regional hydroclimate, including possible effects on the Asian monsoon and southern African precipitation[99–108].

Fires lower surface albedo, via charred land surfaces[109,110], as well as darkening snow and ice surfaces that receive deposition of long-range transported smoke[88,92,109,111–113]. However, severe stand-replacing fires can increase winter albedo by exposing snow and reducing canopy masking[75,76,79,114–116].

These coupled feedbacks highlight the complexity of wildfire-climate interactions and the difficulty of representing them comprehensively in models[3].

Overview of fire modelling frameworks

Wildfire projections are generated using three broad modelling approaches: land surface models (LSMs), Earth system models (ESMs) and statistical or machine learning (ML) models. LSMs and ESMs are process-based approaches which simulate fire occurrence and spread using parameterisations of vegetation, fuels, weather and ignitions, whereas statistical and ML approaches infer relationships between historical fire activity and environmental or socioeconomic drivers. Increasingly, hybrid frameworks combine elements of both approaches.

These three approaches are summarised in Fig. 3, with further framework intercomparisons in section 'Synthesis of modelling approaches'. The number of studies identified for each model type is summarised in Table S1 and all model acronyms are spelt out in Table S2.

Land surface models

LSMs are process-based models that simulate terrestrial vegetation structure, carbon cycling, and land-atmosphere exchanges. Most represent vegetation using plant functional types and include photosynthesis, stomatal conductance, vegetation dynamics, mortality, litter, soil decomposition and surface energy fluxes[117–120]. Some additionally include nitrogen cycling to constrain CO_2 fertilisation responses.

Many LSMs include explicit fire modules that simulate ignitions, fuel availability and moisture, fire spread, human suppression, burned area, fuel consumption and fire emissions[121,122]. These fire parametrisations may be embedded directly within the LSM[118,123,124] or in a fire module that multiple LSMs can use[74,125–129]. LSMs require prescribed meteorology as input, or they can run online, coupled within an ESM (Section 'Earth system models'). Major LSMs and their associated fire modules are summarised in Table S3.

Recent LSM intercomparison projects, including TRENDY and FireMIP[55,122,130–133], demonstrate substantial variability in simulated burned area and fire emissions. Interannual variability is frequently underestimated[122,134,135]. In FireMIP, this was attributed to models that did not capture the fuel buildup resulting from the vegetation productivity of previous seasons[122].

Historical evaluation is typically conducted against satellite-derived burned area and emissions datasets, including GFED, MODIS and FireCCI[119,136–139]. However, these observational products themselves contain considerable uncertainties (up to a factor of 3–5 globally)[140–143]. Advanced benchmarking frameworks, including ILAMB and AMBER[124,144,145], and sensitivity experiments isolating drivers such as CO_2, lightning, or population density[139,144,146], are increasingly used to evaluate model behaviour and uncertainty. Models with improved representation of vegetation dynamics, agricultural burning and human suppression generally perform better against observations[122,132]. The treatment of fuel flammability and moisture is a large source of inter-model variability. Models estimate these quantities using proxies such as fire danger indices, vapour pressure deficit, soil moisture, relative humidity, temperature and precipitation[139,147]. Representation of agricultural burning also varies across models[122,132], or is sometimes not included at all. Finally, biases in the climate models used to drive LSMs propagate directly into simulated fire behaviour[136,148].

Earth system models

ESMs simulate coupled interactions among the atmosphere, ocean, land and cryosphere. Future fire projections using ESMs generally fall

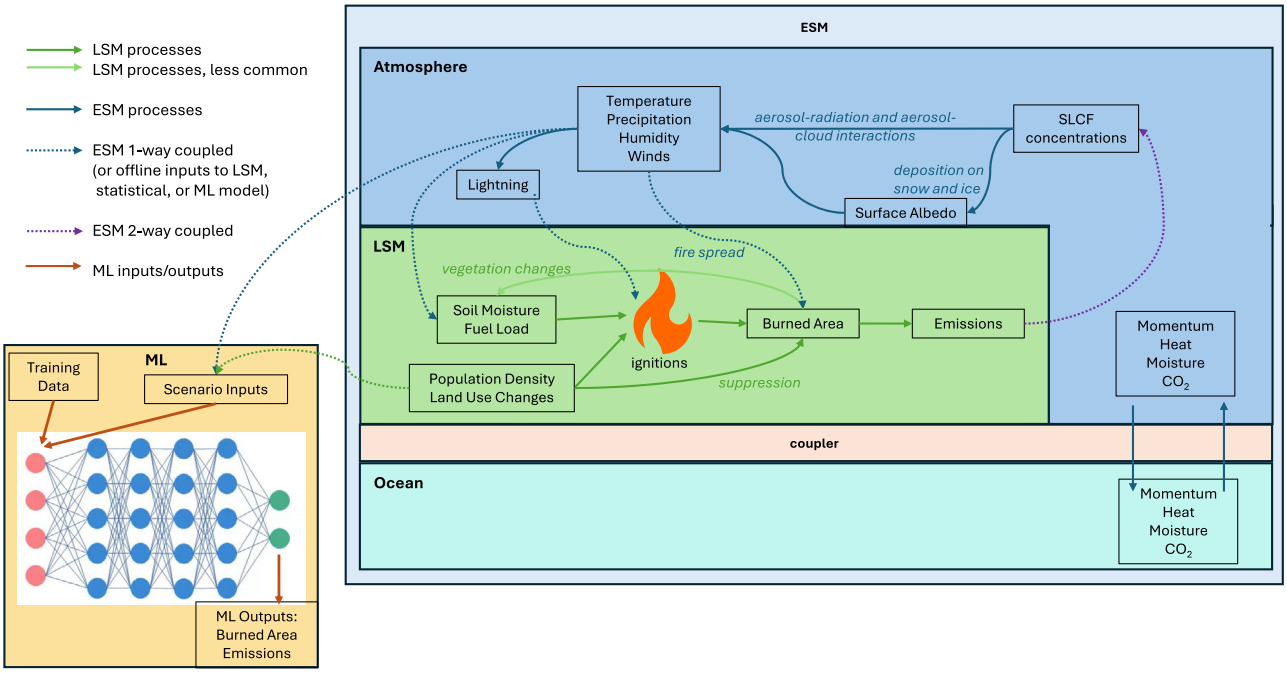


Fig. 3 | Schematic representing the components of the different types of climate-fire models, and the options for coupling between model components. Pale colours indicate less routine processes, and dotted arrows are for less routine couplings/interactions. SLCF short-lived climate forcers.

into two categories: (1) prognostic simulations in which fires are explicitly simulated within coupled land-atmosphere frameworks (Table S3), and (2) studies that use ESM meteorological output to derive fire weather or fire risk indices (SI Note 3). The results from both types of studies are included in section 'Modelling future fires under climate change'.

Prognostic ESM fire simulations may be one-way or two-way coupled. In one-way coupled configurations, simulated meteorological fields drive fire occurrence and burned area, but the resulting fire emissions do not feed back onto the atmospheric system. In two-way coupled models, fire emissions are injected into the atmospheric component of the ESM, allowing interactions with atmospheric chemistry, aerosols, radiation, clouds and climate[128,149–152] (Fig. 3). Most ESMs with prognostic fire modules are still one-way coupled, and model diversity remains limited. For example, in CMIP6, 19 ESMs submitted prognostic burned area or fire CO_2 emissions, but these simulations were derived from only three primary fire modules embedded within nine LSMs[153].

Evaluation of ESMs also includes atmospheric concentrations of fire-emitted species, aerosol optical properties and wet and dry deposition. Community tools such as ESMValTool facilitate these comparisons using satellite and surface observational datasets[154–156]. ESM performance for the historical period has been evaluated for one-way coupled models in CMIP5 and CMIP6 intercomparison studies[153,157,158], which demonstrated improvements relative to earlier model generations, attributed to advances in fire parameterisations and reductions in host-model climate biases[153].

Two-way coupled ESMs introduce additional complexity with explicit treatment of fire-emitted gases, aerosols, plume injection heights, atmospheric chemistry and aerosol-cloud interactions. Plume injection height is a particularly important component of the coupling process because it strongly influences aerosol lifetime, long-range transport and radiative forcing[88]. Emission speciation is derived from observationally constrained emission factors[159,160] or parameterisations[150] although they also remain highly uncertain a variable[140,161–163].

Although ESMs with prognostic fire capability have existed for more than a decade[128,149–153], relatively few studies have applied fully coupled simulations to future wildfire projections.

Statistical and machine learning models

Statistical and ML models provide a data-driven alternative to process-based wildfire modelling. Rather than explicitly simulating physical fire processes, these approaches infer relationships between historical fire activity and environmental or socioeconomic drivers using observational datasets ranging from satellite-derived burned area to demographic indicators. Most statistical and ML frameworks rely on combinations of meteorological, vegetation, topographic and anthropogenic predictors[164–170].

As the availability of geospatial datasets has expanded, wildfire modelling has evolved from traditional regression relationships to ensemble and deep-learning frameworks capable of capturing non-linear interactions among climate, vegetation and human activity[165,167,169,171–175]. These models are increasingly used for wildfire prediction, fuel mapping, fire behaviour analysis and as diagnostic tools for identifying dominant controls on regional fire activity[176]. SI Note 2.1 and Table S4 contain more details on ML methods and types.

Model evaluation is typically conducted against historical satellite-derived burned area and fire emissions datasets. Methods such as SHapley Additive exPlanations (SHAP) analysis are also increasingly used to improve interpretability and identify the dominant drivers of model behaviour[177].

Synthesis of modelling approaches

Figure 3 summarises the three different modelling approaches for simulating or projecting fires in a changing climate, and their inter-connectivity. While both LSMs and ESMs are process-based, LSMs are less computationally expensive than fully-coupled ESMs. Despite continued improvements, both types of models still struggle to reproduce fire seasonality, interannual variability, extreme fire years, human interactions and regional fire distributions[132,136,139,147,178,179]. Important limitations include simplified ignition schemes based on climatological lightning or population density relationships[36,119,120], coarse spatial resolution that under-represents small fires and fine-scale vegetation heterogeneity[120,180,181], and incomplete representation of processes such as peat fires, tropical deforestation fires and multi-day fire persistence[131,182].

Each model type provides a complementary advantage and application. For example, two-way coupled ESMs provide the most complete and

self-consistent representation of fire-climate interactions, allowing for climate-driven changes in fires[183,184] (Figs. 1 and 3). However, due to the chaotic nature of the climate system and the stochastic nature of fire ignitions, they do not reproduce actual historical fires. Additionally, greater model complexity creates more opportunities for biases to accumulate. ESMs inherit limitations from their LSMs and also introduce additional biases in atmospheric variables[48,185–188]. However, ESMs driven by historically constrained BB emissions can well support detection and attribution studies. LSM simulations can be valuable for disentangling drivers of model diversity, and running the same LSM within several host ESMs can provide insights into the relative importance of host model climate[153,157].

Statistical and ML models are fundamentally different from ESMs and LSMs, though ML models are often trained from ESM and LSM output. Compared to process-based approaches, ML models are fast to run, taking advantage of the proliferation of potential input datasets. Comparisons between ML and process-based approaches remain relatively limited, though available studies demonstrate that ML models can outperform process-based models in reproducing historical burned area[189] and $PM_{2.5}$ emissions[190].

Statistical and ML models face several important limitations. Their performance depends strongly on the quality and representativeness of training datasets[175,191]. While projections from all model types are sensitive to biases in the underlying climate models, most ML frameworks do not explicitly enforce physical consistency, limiting their ability to simulate coupled Earth system responses[164,167,192]. These differences become most apparent when simulating compound fire-weather extremes, where fire behaviour depends on the simultaneous evolution of interacting meteorological and fuel variables. As these relationships may change under future climates, ML models trained on historical observations are less reliable when extrapolated beyond the range of past conditions than process-based models that explicitly represent the governing physical processes. Predictive skill also varies geographically, with lower performance often occurring in regions with complex topography or ecological transition zones[193,194].

Increasingly, hybrid ML-process approaches are being used to diagnose deficiencies in process-based fire models and improve future projections[134,176]. For example,[176] used random forests to identify vegetation-fire interactions as a major source of uncertainty in global fire-enabled vegetation models. In addition, process-based and statistical models can be used as complementary tools in the same analyses, for example, in attribution studies linking anthropogenic climate change to increases in large fires[70].

For all modelling approaches, assessing historical performance is a prerequisite for reliable future projections. However, it is critical that models get the right answer for the right reasons, as the drivers of future trends may differ from those in the historical period, and models are tuned to present-day conditions. For instance, LSMs and ESMs sometimes fail to reproduce the observed decline in global burned area since 1850, and this was attributed to fire suppression and land use changes[121,122,129,150,153,195]. However, trends are already changing since the satellite era[134,135,189,195,196], and these trends may continue to change in the future, via policies around deforestation and fire management. At the same time, CO_2 fertilisation leads to enhanced vegetation growth and fuel load, and unprecedented warming will increase flammability. Representing regional differences in dominant fire drivers remains challenging. Evaluation based only on global metrics is dominated by tropical burning and can obscure poor performance in climatically sensitive regions such as boreal forests[88,121,138,181].

Despite limitations, process-based and ML models serve as useful frameworks for future fire projections[135,153]. Modern models extend beyond traditional ESM applications that provide meteorological data for estimating fire risk. They can support climate research and policy by examining the role of fire in future climates[147,150,152], including the effects of changing fire emissions and land use management.

Modelling future fires under climate change

Here, we provide a cross-framework synthesis of future projections of burned area, fire emissions, fire seasonality and fire risk from global and regional fire studies, with emphasis on end-of-century changes under CMIP climate scenarios[183,197,198]. Details on future scenarios are provided in SI Note 4. The results outlined below are broadly consistent with qualitative patterns across all model types, despite differences in methodology, though they do differ substantially in the magnitude of projected changes.

Projected changes in future fire activity

Figure 4 summarises projected future changes in global burned area and fire emissions from the literature across low-, moderate- and severe-warming scenarios. Despite large methodological differences among LSMs, ESMs and ML approaches, most studies project increasing global burned area and fire emissions by the late century, particularly under severe warming scenarios[119,120,139,146,157,158,173,199,200].

The magnitude of projected changes varies greatly, partly due to differences in scenario definitions and averaging periods, but mainly because fire projections depend strongly on scenario choice, climate sensitivity, vegetation dynamics, human-fire interactions and model structure. MLs (e.g. from ref. 173) could project particularly strong increases under severe warming scenarios but may be sensitive to extrapolation beyond historical climate states. In contrast, some LSMs project weaker global increases or even regional declines where anthropogenic suppression, fragmentation, or reduced fuel availability offset climate-driven increases in fire weather[144].

Global fire carbon emissions are commonly projected to increase by roughly 50–100% under severe warming scenarios[157,174,199,200]. Lower-warming scenarios produce smaller changes or near-stabilisation of fire activity by late century[119,158], although some studies still project increasing fire emissions[201]. Idealised CO_2 experiments similarly suggest strong sensitivity of burned area and fire emissions to warming and CO_2 fertilisation[35,202].

Coupled ESMs demonstrate that fire-emitted aerosols can influence regional temperature and precipitation through radiative and cloud effects[88], highlighting an important advantage of fully interactive land-atmosphere modelling frameworks. However, few studies simultaneously include fully coupled fire aerosols, atmospheric chemistry and greenhouse gas feedbacks.

Regional projections are summarised in Fig. 5. Across modelling frameworks, the clearest global signal is a strong poleward expansion and intensification of wildfire activity under climate warming[4,120,200,203,204]. These regions are projected to experience much longer fire seasons, more severe fire weather, increased lightning ignitions and large increases in burned area and fire emissions[8,21,35,48,139,205–211]. Burned area and fire carbon emissions may double or more under severe warming[139,193,204,205,208,212].

In Canada, the annual burned area could increase to nearly five times the modern historical average by the 2090s under severe warming[139]. In China, a random forest ML approach projected increasing fire occurrence across 60–70% of forested regions by the late century, with large increases in burned area and an expansion of moderate- and high-risk zones[213]. Mediterranean ecosystems and western North America also emerge as major hotspots of increasing fire danger, with projections consistently showing more frequent heatwave- and drought-driven fire weather, larger fires and expanded fire seasons[35,171,214–216]. Southern Europe and the western United States may be especially vulnerable to increases in regional fire activity that could strain suppression capacity[217–219]. Under severe warming, burned area increases exceed 100% across parts of southern Europe[165,171], and increases are significant even in a moderate warming scenario[165,220].

Projections for tropical regions are generally weaker, mixed, or declining[120,137,167]. Tropical and subtropical regions exhibit much greater divergence among projections and show greater regional variability. In parts of the Amazon, Cerrado and subtropical Africa, warming and drying are projected to increase fire probability, fire weather severity and ecosystem vulnerability, particularly where climate change interacts with deforestation and fragmentation[35,199,221–226]. However, most models project stable or declining burned area across portions of tropical Africa and South America because of reduced fuel availability, woody thickening, increasing precipitation, or enhanced human activity[36,137,144,167,199,200,202,225,227,228]. The changes can be mixed, with declining burned area across parts of northern

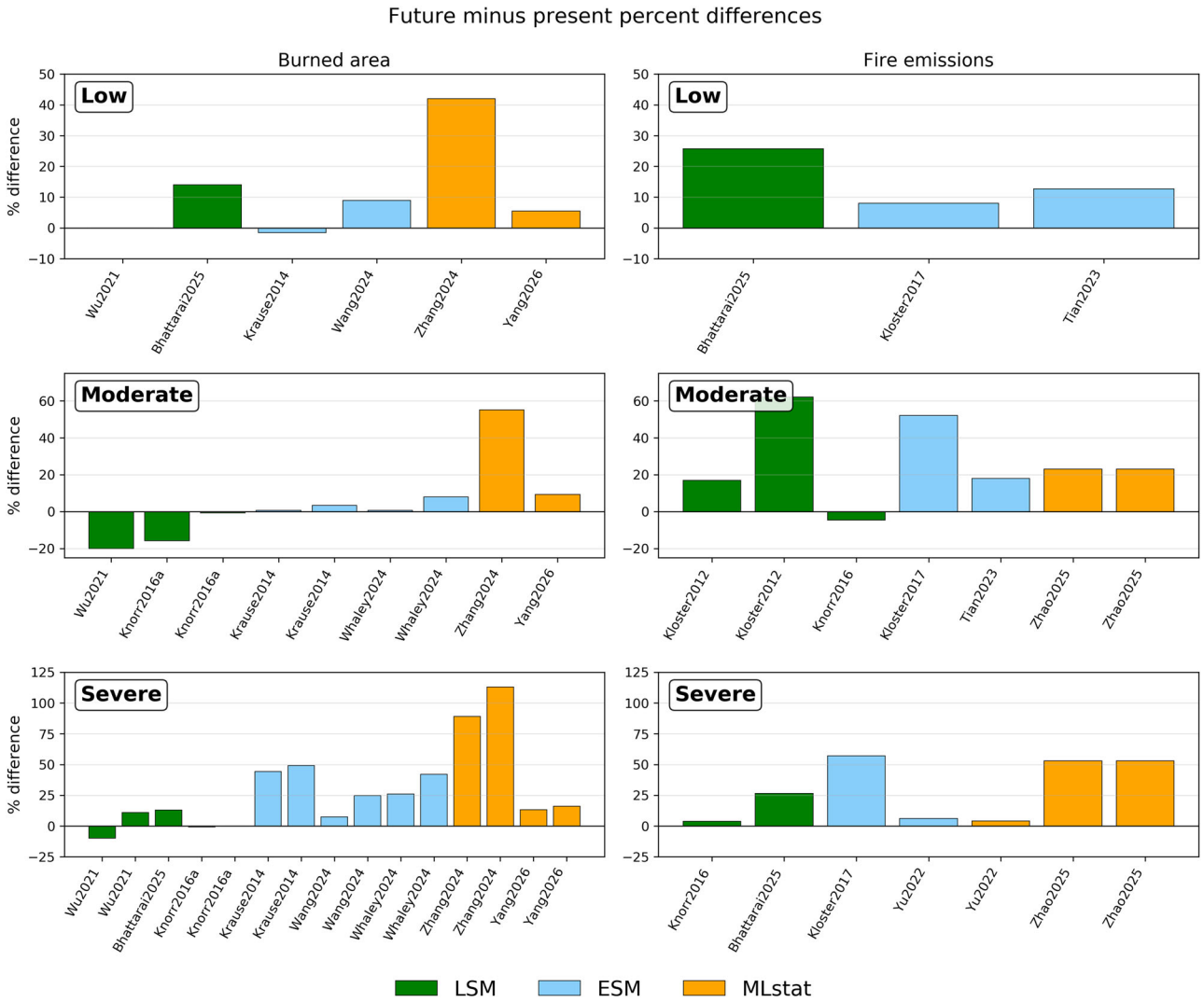


Fig. 4 | Future vs present changes in fire activity. Bars indicate the projected percent change in global burned area (left) and fire emissions (right) relative to present-day conditions across low-, moderate- and severe-warming scenarios, synthesised from quantitative results in the literature (see Table S6 for details).

hemispheric Africa but increasing fire activity across southern tropical South America under severe warming as in Li et al.[167]. The Congo Basin represents an example of methodological divergence, with some ESMs projecting declining fire activity associated with increased precipitation or reduced lightning ignitions[48,88,229], while observation-constrained and ML approaches instead project enhanced fire emissions under intensified drying[174,199]. Even within individual studies, the projected sign of tropical fire changes can depend on methodological approach. For example, Yu et al.[199] projected declining tropical fire emissions using direct ESM output, but increasing emissions when constraining the same simulations using ML techniques. The relative importance of hydrological sensitivity, CO_2 fertilisation, vegetation change and human activity, therefore, remains highly uncertain across tropical ecosystems.

Changes to fire behaviours and impacts. Fire seasons lengthen across northern mid-latitudes. Boreal and temperate fire seasons are projected to lengthen considerably[35,48,119,226,230], often by one to 3 months[21,137,148,206,211,214,231,232]. In Fennoscandia, for example, the fire season may begin several weeks earlier and end up to 1 month later under severe warming[211]. Fire intensity also increases globally and regionally, from LSM and ML-based models. For example, using JSBACH-SPITFIRE, Veira et al.[137] projected a 10–50% increase in the fraction of high-intensity fires under severe warming scenarios, and Dong et al.[230], using a random forest ML model, simulated increases in large-fire days in southern California. Increasing biomass and burned area contribute to stronger smoke-related air quality impacts[180,233,234]. Several studies project increasing fire intensity and smoke exposure, such as large increases in wildfire $PM_{2.5}$ concentrations and smoke-related health risks across boreal regions, the western United States, Latin America, Africa and Asia[164,180,192,233–235].

Many future wildfire studies derive fire weather or fire risk indices offline using ESM meteorological output rather than explicitly simulating fires. These approaches represent one of the earliest and most widely used methods for projecting future wildfire change[50,236]. Fire risk studies using ML models also exist. Most studies project pronounced increases in fire weather severity, fire-prone area and extreme fire conditions by the end of the 21st century[4,21,219,237,238]. Fire-prone conditions are projected to expand across much of the global burnable land surface[238], with particularly large increases in boreal and temperate regions[4,219]. Extreme fire weather days are projected to double or triple in some regions[21,217,219], while the duration of rare extreme fire events may increase considerably under severe warming[232]. Transitions toward higher fire-danger classifications will increase[206,207,239], and significant increases in fire probability and fire risk are projected in Canada, China, Russia, the Mediterranean, the Alps, the western United States, and the Amazon[170,213,218,222,240,241]. Although fire-weather and fire-risk approaches focus primarily on

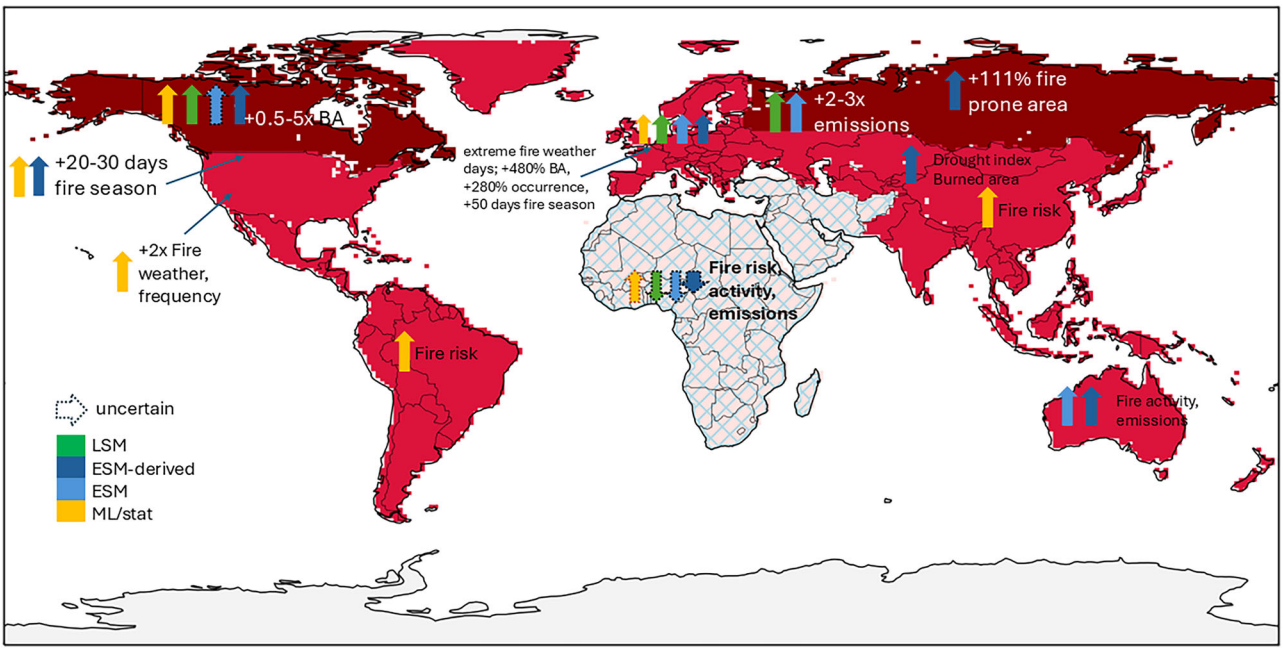


Fig. 5 | Schematic synthesis of projected regional wildfire changes across modelling frameworks. Colours indicate the approximate magnitude and confidence of projected change, while arrows indicate modelling framework type (LSM = green, ESM = light blue, ESM-derived = dark blue, ML/statistical = orange). Dotted outlines indicate regions with particularly large inter-model uncertainty or disagreement (see Table S6 for details).

meteorological drivers, several studies also identify important roles of land use, fuel connectivity and human activity[215,222,240,242].

Dominant drivers of fire changes

Projected fire changes arise from interacting climatic, ecological and anthropogenic drivers, with projected fire change values differing considerably among regions because the dominant controls on fire activity vary geographically (Fig. 2). Several studies identify warming[21,23,214,225,237,243] and drying as primary drivers of increasing fire activity[119,202,226,233], while others emphasise the importance of CO_2 fertilisation and increasing fuel availability[35].

In high-latitude regions including Canada, Siberia and Fennoscandia, increasing temperature, vapour pressure deficit and declining fuel moisture are consistently identified as dominant drivers of future fire increases[4,35,120,139,158,202,207,211,229,237], with lightning ignition playing a significant role there as well[139]. Boreal regions are generally projected to remain moisture-limited, with warming-driven fuel drying dominating over fuel constraints[211].

Semi-arid regions such as the southwestern United States, the Brazilian Highlands, Australia and Central Asia are also projected to experience increasing fire risk associated with drying and increasing atmospheric moisture demand[118,229], however, future fire activity remains strongly fuel-limited, making biomass availability and vegetation change particularly important[35,202,239,244]. Some studies emphasise the importance of fuel-limited versus moisture-limited fire regimes[136,146]. In parts of the Mediterranean, climate-driven drying may be partially offset by reduced net primary productivity and declining fuel availability[136,146], and precipitation and CO_2 fertilisation can increase fire risk by enhancing fine-fuel production[202,216,237,239,244,245].

Drivers in tropical regions are complex and result in differing future projections. Moist tropical regions remain especially sensitive to precipitation variability and drought occurrence[216,229,245]. In the Congo Basin, declining fire risk could be due to projected increases in precipitation[229], whereas in tropical South America, there may be increasing fire activity and vegetation transitions under warming and drying[226]. In Mediterranean climates, Amazonia and southern Africa, future fire risk is strongly influenced by combined warming and drying trends[215,246]. Interpretable ML frameworks indicate that precipitation and vapour pressure deficit are dominant controls on tropical burned area variability[167]. Human influences remain important in some regions, increasing fire susceptibility in Chile[247].

Vegetation shifts also influence future fire behaviour. Increased CO_2 can enhance vegetation productivity and fuel loads, particularly at high latitudes[36,120,146], while woody encroachment in tropical savannahs may reduce grass-fuel continuity and suppress fire spread[36,227]. However, some LSMs simplify vegetation-fire feedbacks, potentially exaggerating some of these responses[146], though inclusion of dynamic vegetation changes amplifies projected future burned area in the United States Liu and Wimberly[192].

Overall, the synthesis suggests a future global fire regime characterised by strong high-latitude amplification, intensifying fire-weather extremes across many temperate regions, and more uncertain but potentially transformative changes across tropical ecosystems[4,120,203].

Highlighting key uncertainties

Future fire projections remain highly uncertain across all modelling frameworks because fire activity depends on interacting climatic, ecological and anthropogenic processes that are themselves difficult to project under climate change. The dominant uncertainties limiting confidence in future projections are structural rather than parametric. Differences in future climate forcing and in the representation of vegetation, human activity and fire processes generally contribute more to projection spread than uncertainty in individual parameter values. Parameter uncertainty remains important, particularly for emissions, combustion and atmospheric chemistry, but typically has less influence on projected long-term changes in fire activity than structural differences among modelling frameworks (Table 1).

One of the largest sources of uncertainty originates from the climate models used to drive wildfire projections. LSM can exhibit up to a factor-of-five range in future fire emissions when driven by different climate models because differences in projected temperature and precipitation strongly influence vegetation productivity, fuel moisture and fuel availability[148]. Although ML approaches have improved historical fire simulations and attribution capabilities, they are also subject to climate forcing uncertainties. Divergence among driving ESMs remains a major limitation, with some studies showing disagreement in the direction of future fire-risk changes

Table 1 | Sources of key uncertainties in fire modelling

Source of uncertainty	Type	Relative importance	Why
Climate projections	Cross-cutting	Very high	Propagates through every framework
Vegetation/fuels	Structural	Very high	Controls fuel availability
Human-fire interactions	Structural/Parametric	High	Dominant tropical and populated regions
Fire process representation	Structural	High	Spread, ignitions, plume height
Lightning	Structural/Forcing	Moderate	Important in remote boreal regions
Observational constraints	Structural	Moderate	Limits evaluation and ML training
Numerical parameter choices	Parametric	Lower	Tuning for historical agreement

across more than half of land areas during the near term[245]. Consequently, ensemble weighting and Bayesian averaging approaches are increasingly used to reduce multi-model bias and improve projection robustness[158,238].

Projected fire responses may also strongly depend on the representation of moisture limitation and fuel drying. Allen et al.[228] found that shortwave warming associated with aerosol reductions enhanced fuel drying more strongly than longwave greenhouse-gas warming. However, the sensitivity of these responses depends strongly on how fuel moisture constraints are represented, particularly in boreal regions where relative humidity may be a more important control on fire activity than soil moisture alone[138,139].

Lightning ignition is also a large uncertainty for fire projections, as different lightning parameterisations provide diverging projections[185], and some models do not represent historical trends as compared to measurements[44,47,50,210,225], greatly reducing confidence in their future projections. There is also no consensus on how much changing lightning will impact changes in global burned area. Its impact may be modest relative to fuel and climate effects[200,202], or may be significant[48,210,248]. However, there is general agreement that lightning is a key driver of boreal fire variability[120,139].

Human activity introduces uncertainty and competing fire-influence that differs regionally and over time. There is parametric uncertainty through regionally varying impacts of population density on fire ignition and suppression[144,148,153,157,179,227]. These influences are particularly important in tropical and densely populated regions[36,137,179,227], where factors such as agricultural expansion, urbanisation, landscape fragmentation and fire suppression have historically reduced burned area in many regions[57], whereas escaped fires increasingly dominate burned area during extreme fire seasons[249].

Additional uncertainty arises from the historical datasets used to train and evaluate ML models. Satellite and reanalysis products can exhibit conflicting trends and extremes in fire-weather indices, propagating uncertainty into bias-correction and prediction frameworks[250]. Furthermore, many ML approaches rely on static vegetation and anthropogenic predictors that do not evolve dynamically with climate change, limiting their ability to capture future changes in fuel structure, vegetation composition and human infrastructure[133,203]. Although hybrid ML-process frameworks may help address some of these limitations (Section 'Integrating AI with process-based models'), uncertainty increases under severe scenarios where future climate states diverge most strongly from the historical training record[243]. Regional sensitivities also vary. For example, projections across the European Alps are strongly influenced by uncertainties in lightning occurrence and precipitation seasonality[170].

Collectively, these uncertainties highlight the importance of multi-model ensembles, high-resolution downscaling, improved representation of dynamic vegetation and human-fire interactions, and tighter integration between process-based and data-driven modelling frameworks[133]. Uncertainties can also be well addressed via intercomparison projects.

Insights from intercomparison studies

Older intercomparison studies based on derived fire-weather indices revealed large uncertainties during the near term. Using fire indices derived from 16 ESMs, Moritz et al.[245] found weak model agreement for projected fire activity changes during 2010–2039, with agreement strengthening by the late century as warming signals intensified. Bowman et al.[178] emphasised the importance of dynamic vegetation-fire interactions and argued that fire-weather approaches alone are insufficient to capture coupled fire-climate feedbacks.

At the time of writing, the only multi-model intercomparison study of future prognostic fires was Kloster and Lasslop[157], who compared six CMIP5 ESMs with prognostic fire capability (nine model configurations total) for both the historical and future time periods. Their results demonstrated a divergence in projected future fire carbon emissions across models and scenarios (Fig. 6a–c), despite broad agreement toward increasing fire activity under stronger warming scenarios. This large inter-model spread reflects differences in climate sensitivity, vegetation responses, fire parameterisations and anthropogenic influences among ESMs.

No fully published intercomparison of prognostic future fires from CMIP6 ESMs is yet available. However, emerging analyses from CMIP6 models highlight an increasingly important distinction between prescribed and prognostic future fire emissions[251–253]. Prescribed CMIP BB emissions primarily reflect land-use changes, whereas prognostic simulations respond dynamically to changing climate, vegetation and weather conditions. Consequently, future fire emissions differ depending on whether fires are prescribed or interactively simulated (Fig. 6d–f).

Intercomparison studies demonstrate the importance of ensemble approaches for quantifying projection uncertainty (e.g. ref. 254). For example, Cane et al.[255] compared projections for an alpine region from a single regional climate model and a seven-model ensemble and found that the multi-model ensemble projected increases in fire potential and extreme fire-weather days, while the single-model simulation projected weaker or even opposite regional trends because of differences in precipitation responses. Thus, projections are sensitive to regional hydroclimate change, and multi-model approaches are generally more robust than individual simulations, particularly in regions with complex terrain and precipitation dynamics.

Emerging opportunities and path forward

While future projections of fires are highly uncertain, given the strengths of each modelling approach, we summarise in Fig. 7 their best use cases. There are also several avenues that are recommended for a path forward, given how important reliable future fire projects are to society.

Gaps in process-based models

One of the largest remaining gaps in wildfire modelling is the limited representation of two-way fire-climate feedback within ESMs (Section 'Wildland fire controls on climate'). Although some studies now include fire-emitted aerosols and greenhouse gases[256], most fire projections still neglect interactive impacts of chemistry, smoke-cloud interactions and atmospheric oxidative capacity. This omission may lead to systematic underestimation of long-term climate feedbacks from future fires. For example, Chen et al.[86] incorporated gas-phase chemistry and found that future fire emissions substantially reduced atmospheric oxidative capacity, increasing methane concentrations to a similar magnitude as increasing wetland methane sources, and direct fire-related CO_2 emissions.

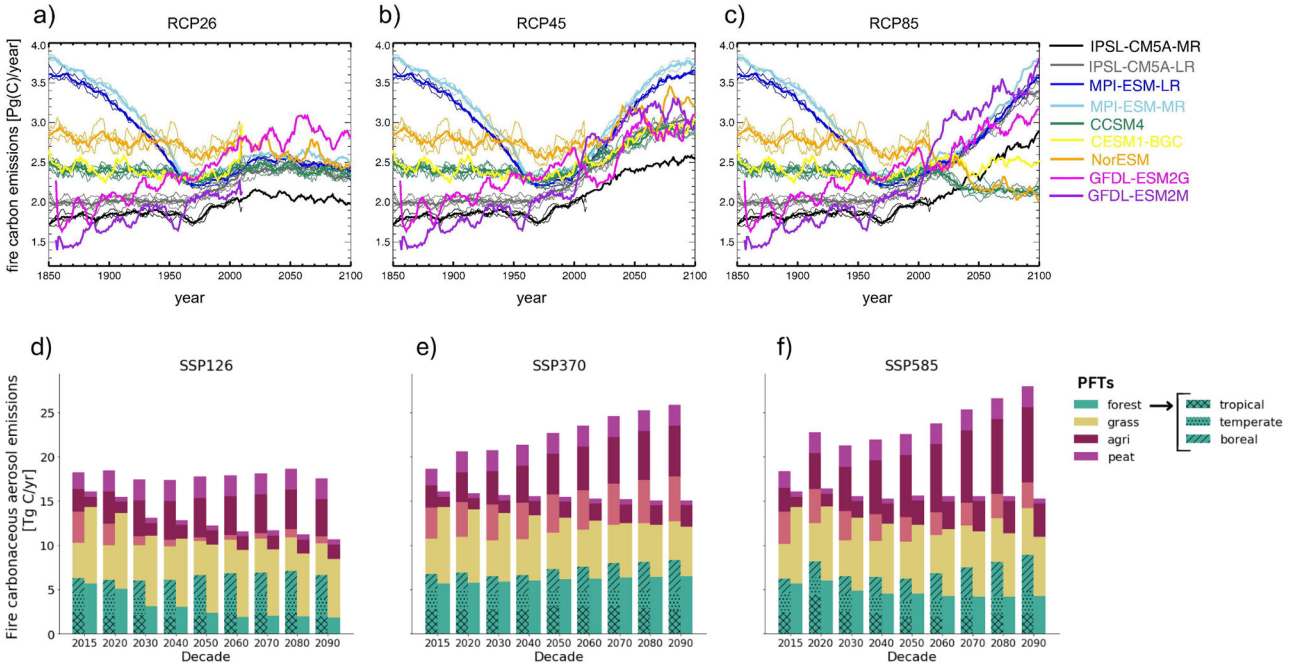


Fig. 6 | CMIP5 and CMIP6 global fire emissions. (top row): Multi-model comparison of future fire carbon emissions from CMIP5 ESMs with prognostic fire capability under three climate scenarios: **a** RCP26 (low), **b** RCP45 (moderate) and **c** RCP85 (severe), from Kloster and Lasslop[157] with permission. (bottom row) Projected future fire carbonaceous aerosol emissions, multi-model decadal averages from CMIP6 ESMs with prognostic fire capability, comparing simulations with climate-responsive fire feedbacks (left bars) and prescribed BB emissions (right bars). **d** For SSP126 (low), **e** SSP370 (severe) and **f** SSP585 (severe), from Whaley et al.[252] with permission.

Current ESMs also incompletely represent longer-term cryosphere feedbacks associated with wildfire aerosols. While many models simulate black carbon deposition onto snow and ice surfaces[257], they often neglect multi-year accumulation and persistence of deposited aerosols within Arctic snowpacks and glaciers. These long-memory effects may become increasingly important as accelerated polar warming exposes older snow and ice layers containing previously deposited fire-derived aerosols.

Another major limitation is the treatment of fire persistence. Many LSMs and ESM fire modules simulate fires independently at daily timesteps and therefore cannot fully represent prolonged fire spread associated with large remote wildfires[122]. However, fire duration is a major control on burned area in regions such as boreal Canada[212]. Earlier approaches attempted to approximate persistence by linking fires across consecutive days[258,259], while more recent studies have suggested using satellite-derived fire duration datasets to develop explicit fire-duration parameterisations[132,260]. A small number of LSMs now include multi-day fire behaviour[182,261,262], however, these approaches have not yet been applied in future projection studies.

Finally, peat fires are another important but incompletely represented process in global fire models. Peatlands have unique hydrological controls on fire occurrence and distinct fire-emission speciation[263], while storing long-term carbon reservoirs that are increasingly vulnerable to warming and drying[31,264,265]. Despite peat fires being especially important in northern boreal and Arctic regions, which are warming faster than the global mean, they are not included in all global BB inventories[140], and relatively few LSMs explicitly simulate peat fires. Notable exceptions include the Li fire scheme[78,128,266] and INFERNO-peat[267]. Including and improving peat-fire representation will be essential for accurately projecting future carbon-cycle and climate feedbacks associated with high-latitude wildfire activity.

Frameworks for coordinated multi-model studies

The large diversity in fire metrics, regional definitions, scenarios, averaging periods and modelling frameworks currently limits direct comparison among future fire studies. Coordinated multi-model intercomparison frameworks are therefore essential for identifying robust emergent patterns, quantifying uncertainty and separating structural model differences from scenario-driven variability.

Community efforts are increasingly addressing these challenges. FireMIP Phase 2 extends earlier historical intercomparisons toward future fire projections using multiple LSMs under consistent forcing scenarios[133]. Such coordinated experiments will provide important insight into the sources of uncertainty associated with vegetation dynamics, fuel representation, climate sensitivity and human-fire interactions across next-generation fire-enabled LSMs.

The HTAP3 Fires project[252] provides another framework for coordinated future fire experiments, including simulations using both prognostic fires from ESMs and prescribed future BB emissions that are climate sensitive[253]. HTAP3 Fires includes standardised regional definitions based on fire-biome characteristics, enabling more consistent comparison of burned area, emissions, atmospheric composition and fire-related climate impacts across modelling systems.

Future CMIP7 activities are also expected to expand coordinated fire simulations. AerChemMIP2 will encourage ESMs with prognostic fire capability to provide historical and future scenario simulations[268]. CMIP7 additionally emphasise emissions-driven simulations, particularly for CO_2, allowing more self-consistent representation of carbon-cycle feedbacks and improved assessment of Earth system responses and potential climate tipping points[184,269,270].

These coordinated project provide an important pathway toward more robust, comparable and policy-relevant future fire projections.

Integrating AI with process-based models

Hybrid approaches that integrate ML with process-based models are a promising direction for improving future fire projections and reducing computational limitations. For example, Kavhu et al.[134] developed a statistical burned-area model designed for coupling with LSMs, using generalised linear models to represent relationships among biophysical and socio-economic drivers simulated within process-based frameworks.

☑ prognostic ☑ prescribed ☐ missing/none ☑ data-driven/trained	LSM	Typical ESM (without interactive fires)	ESM with interactive fires	ML/statiscal models
Strengths	separates drivers; same model can use different forcing datasets to quantify forcing uncertainty.	coupled climate, well-vetted, simulates transport of prescribed fire emissions	resolves smoke transport and two-way climate-chemistry-ecosystem feedbacks.	extremely fast after training and suitable for large ensembles and rapid projections.
Limitations	cannot simulate smoke transport or two-way atmospheric feedbacks; sensitive to forcing and fire/fuel	lack of fire-on-climate feedback and fire emissions not consistent with climate, thus future projections are limited to fire weather	costly; coupled and compensating biases plus internal variability obscure driver attribution.	only as reliable as the training data; domain shift, spurious relations, and physical inconsistency remain risks
Shared challenges	*imperfect burned-area, emissions, fuel, and human-activity data; meteorological forcing bias; scale mismatch and unresolved extremes; sparse validation; structural uncertainty*			
Best use cases	vegetation transitions, fire process development	seasonal and century-scale fire weather predictions and projections	global climate feedbacks, century scale projections	seasonal fire danger, operational prediction, uncertainty diagnoses
Components:				
atmospheric dynamics and chemistry	☑	☑	☑	☑
vegetation dynamics	☑	☑	☑	☐
fuels (load, moisture, continuity)	☑	☑	☑	☑
fire processes (ignition, spread, intensity)	☑	☐	☑	☐
emissions (gases, aerosols)	☑	☑	☑	☑
climate feedbacks (radiation,clouds, circulation)	☐	☑	☑	☐

Fig. 7 | Summary of modelling approaches. Structural components, and key take-away messages on their strengths, limitations and best use cases.

Fire plume rise and in-plume atmospheric chemistry, which forms secondary organic aerosol and ozone and involves non-linear interactions that are computationally expensive to simulate using traditional parameterisations, can be replaced with ML approaches that provide efficient surrogate frameworks trained on high-fidelity simulations or observations[172,207]. Such approaches could reduce computational costs while retaining sensitivity to evolving meteorological and fire conditions. Faster emulators are also highlighted for their potential in long-term planning and identifying locations with higher future fire danger[162,271].

Given that process-based models exhibit systematic biases associated with fire emission factors, plume injection heights, atmospheric chemistry and aerosol processes, residual-learning frameworks can identify persistent discrepancies between model output and observations and apply statistical corrections to improve air-quality forecasts[158,172,207,230,248].

Another emerging direction is physics-informed ML, which embeds physical constraints directly within ML architectures[272]. Unlike purely black-box AI approaches, these hybrid frameworks incorporate conservation laws, chemical stoichiometry and biophysical constraints into neural-network training. Such approaches may help preserve physical consistency while improving predictive capability under future climate conditions. Hybrid ML-physics frameworks can accelerate computationally intensive components of atmospheric chemistry models[174], enabling rapid sensitivity analyses of fire-driven atmospheric composition changes.

AI-based inference approaches may also improve representation of poorly observed fire processes, including fuel moisture variability, ignition probability and human-fire interactions derived from multidimensional observational datasets[248,272]. By combining physically-based process understanding with adaptive data-driven inference, hybrid frameworks provide a potentially powerful pathway toward more flexible, computationally efficient and observationally-constrained fire projections under future climate change.

Toward actionable fire-climate projections

As wildfire models become increasingly complex, improving interpretability and societal relevance is becoming an important priority. Explainable AI frameworks, including SHAP, are being used to diagnose why models predict heightened fire activity and to identify dominant environmental and anthropogenic drivers[207,248]. Such approaches may improve confidence in wildfire projections and help translate increasingly complex modelling systems into information that is more actionable for decision-makers and fire managers.

There is also growing demand for high-resolution, user-oriented fire projections at spatial scales relevant to local adaptation and management rather than global or continental averages[133,158,172,181,250]. Future wildfire assessments will increasingly require improved representation of extreme fire behaviour, including prolonged fire-conducive weather patterns, compound extremes and the heavy-tailed nature of rare fire seasons that are often smoothed within monthly or climatological averages[165,175,273].

Human-fire interactions remain a research priority and opportunity. Ignition sources, fire suppression, land management and socioeconomic development all strongly influence future fire trajectories and could be better represented in models[73,224,245,274,275]. For example, incorporating the Human Development Index into an LSM may better represent suppression capacity over time[195,276]. Scenario-based fire-management pathways were developed for Arctic states[277] and could provide a promising framework for future modelling experiments that provide policy-relevant results.

Despite marked advances in wildfire modelling over recent decades, major uncertainties remain in projecting future fire activity. There is broad agreement across model frameworks that climate change will increase fire risk and extreme fire activity in many regions, particularly in the extra-tropics and boreal forests, although large variations persist in the magnitude of these changes. Future progress will depend on improved representation of coupled fire-climate-human interactions. Each model framework provides

complementary information, though emerging hybrid approaches that integrate process-based modelling, observations and explainable ML provide a promising path toward more robust and actionable fire projections across spatial scales relevant to both Earth system science and decision-making.

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Author contributions

All authors contributed to data gathering for the synthesis and co-wrote and revised the manuscript. Y.L. conceptualised the literature review topic and outline, and co-wrote sections 'Statistical and machine learning models' and 'Integrating AI with process-based models'. I.V. created Figs. 1 and 2 and did the bulk of the literature search and organisation/notation of the reference library for the project. I.V. and S.K. co-wrote section 'Mechanistic changes in fires under climate change'. R.D. created Fig. 3, co-wrote sections 'Earth system models' and 'Synthesis of modelling approaches', and edited the complete manuscript. CHW coordinated the literature review and created Figs. 5–7, wrote much of the manuscript and edited the complete manuscript.

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Competing interests

The authors declare no competing interests.

Additional information

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